DOCKER - SIMPLE CHEAT SHEET

Mac + VS Code | Build and run a Python app

1. Create a Dockerfile

In your app folder, keep Dockerfile, app.py and requirements.txt together. Copy this class example:

```
# Base image
FROM python:3.9
WORKDIR /app
COPY . /app
RUN pip install -r requirements.txt
EXPOSE 5000
CMD ["python", "app.py"]
```

Instruction	Meaning
FROM	Base image (Python 3.9).
WORKDIR	Working folder inside the container.
COPY	Copy project files into the image.
RUN	Install dependencies while building.
EXPOSE	Document the app port.
CMD	Command that starts the app.

2. Add a .dockerignore file

This stops local environment and Git files being copied into the image.

```
bmi-env/
.venv/
__pycache__/
*.pyc
.git/
.DS_Store
```

3. Build and run

Run these from the folder containing your Dockerfile:

```
docker build -t bmi-calculator .
docker run -d --name bmi-app -p 5000:5000 bmi-calculator
docker ps
```

Open http://localhost:5000 if the app listens on port 5000.

Remember: docker build creates an image. docker run creates and starts a container. -p 5000:5000 maps Mac port 5000 to container port 5000; EXPOSE alone does not publish it. The app must listen on 0.0.0.0 inside the container.

DOCKER - RUN, CHECK, FIX

Commands for your Docker Desktop session

4. Run the image from Docker Hub

Image from your session: tweakster24/lappy.

```
docker pull tweakster24/lappy
docker images
docker run -d --name lappy -p 8501:8501 tweakster24/lappy
docker ps
docker logs lappy
```

Port 8501 is an assumption, not verified for this image. If http://localhost:8501 does not open, check logs and confirm the app port.

5. Commands to remember

Command	Purpose
docker --version	Check Docker CLI version.
docker ps / docker ps -a	Show running / all containers.
docker logs NAME	Show app output and errors.
docker stop NAME	Stop a container.
docker start NAME	Start a stopped container.
docker rm NAME	Remove a container.
docker rmi IMAGE	Remove a local image.

6. Fix: zsh says “docker: command not found”

You confirmed Docker Desktop and the CLI exist. Add its command folder to zsh PATH once:

```
echo 'export PATH="$HOME/.docker/bin:$PATH"' >> ~/.zshrc
source ~/.zshrc
docker --version
docker ps
```

If Docker cannot connect to the engine, open Docker Desktop, wait until it starts, then retry docker ps.

7. Quick troubleshooting

Problem	What to try
Container stopped	docker ps -a, then docker logs NAME
Name already used	Use another --name or remove the old container if safe.
Port already used	For example: -p 5001:5000; open localhost:5001.
App will not open	Check logs and verify the container port.

Image = template. Container = an instance of an image. Removing a container does not automatically remove its image.